YM4O1 – Assembly of the four-dimensional action-density theory

Model-specific: L

Lattice Gibbs laws and reflections

Wilson laws, positive crossing kernels, chessboards and coordinate/diagonal mirrors in every periodic Gibbs state.

Required: Transfer from adjacent layers to arbitrary finite stacks.

YM4O1: diagonal mirrors and Gibbs transfer

Same position in YM3: YLO1; QED3: LPM1 / LPO1

Model-specific: M

Constrained linear block estimates

On specified regular field pairs: uniform constrained inverses, regularity, exponential decay and local field dependence.

Required: Constants uniform in volume and number of scales.

YM4O1: propagators and local dependence

Same position in YM3: YPO1; QED3: MBM1 / MBO1

Model-specific: G

Nonabelian backgrounds

Symmetric group averages, gauge normalization, constrained critical backgrounds and their local nonlinear response.

Required: Real and complex domains compatible with iteration.

YM4O1: averaging; background and response

Same position in YM3: YBO1; replaces the fermion position in QEO1

Model-specific: E

Marginal action and its coefficient

Gaussian normalizers, subtraction of the quadratic curvature term and the four-dimensional one-loop coefficient.

Required: Uniform coefficient and remainder estimates.

YM4O1: marginal extraction and one-loop sum

Same position in YM3: YEO1; QED3: RNM1 / RNO1

all three

free covariances

E, N, T, H and source variations: one compatible induction. Previous-level bounds control the next step.

Model-specific: N

Exact local nonlinear step

Localize the fluctuation integral, construct its convergent connected expansion and bound the next local activity. Close this step jointly with the history and flow estimates.

YM4O1: fluctuation integral and local expansion

Same position in YM3: YNO1; QED3: NLM1 / NLO1

Model-specific: P

Free Gaussian curvature term

Curvature projections and Wick contractions give a positive leading action-density coefficient with kernel $|x-y|^{-8}$. The interacting error estimate belongs to V.

YM4O1: free curvature term and normalization

Same position in YM3: YGO1; replaces QED3's quartic probe

free curvature coefficient

one bare sequence

all inputs

periodic-state source transfer

periodic-state reflections and chessboards

Model-specific: C

Continuum limits and spectral cones

Compactness: one subsequence for all separated correlations, with the reference-state mean subtracted.

Spectral inputs: a lattice cone for clustering; the constant-one continuum cone for rotations.

YM4O1: separated limits; reflection spaces; cones

Same position in YM3: YCO1; QED3: CEM1 / CEO1

Model-specific: T

Running coupling and first passage

One coefficient sequence on compatible tori; logarithmic asymptotic freedom and a terminal reference-coupling window.

Required: The common induction with N, H and the sourced estimates.

YM4O1: closed flow and first-passage choice

Same position in YM3: YTO1; replaces QED3's common tuning

Model-specific: R

Sources and normalized response

Early: source variations.

Freeze the real choices, transport each source as a local coupling change and retain the exact sourced mass identity. These estimates participate in the common RG induction.

YM4O1: source transport and complex sources

source derivatives

Later: uniform response.

Use the history bounds and chessboards to obtain local energies and one normalizing function for periodic Gibbs states and compatible tori. Control differences of means and factorial bounds on separated cumulants; Z supplies the flat-test derivative bound.

Required: The state-uniform quantitative source estimates.

YM4O1: local energies, normalizer and graded bounds

Same position in YM3: YSO1; QED3: RRM1 / RRO1

lattice cone; limits

flow hypothesis within the same induction

uniform separated bounds

Model-specific: K

Components and clustering

Use the lattice cone and reflected moment bounds to control the plateau. Select pure lattice states and obtain a common continuum phase with inverse-square clustering.

Required: Gibbs-decomposition applicability and the spectral filter.

YM4O1: phase selection and clustering

Same position in YM3: YKO1; QED3: ECM1 / ECO1

Model-specific: H

History measures and partition bounds

Exactness: retain stored fields and their conditional measures, with the exact all-level integral identity.

Stability: large-field suppression, uniform masses and live-lineage age tails summable with the factor $L^{8\alpha}$ from two source derivatives (age α). Close these estimates jointly with the running flow.

YM4O1: stored variables and large-field histories

Same position in YM3: YHO1; QED3: TPM1 / TPO1

real history bounds

history stability and ages

Continuum cone, selected phase and reflected moment bounds for Z

passage to limits and the selected phase

source response

all inputs

Model-specific: V

Two-point law and transfer to the phase

Follow both terms of the sourced covariance identity through the two observation scales. Compare with P and transfer the two-sided connected two-point bounds between Gibbs states and tori, then to the selected phase.

Required: The two Gaussian recursion estimates and accumulated error control.

YM4O1: two-point calculation and state transfer

Same position in YM3: YVO1; replaces QED3's component cumulants

structural inputs and component nontriviality

factorial growth and means

all inputs; common parameters and phase

Model-specific target: Z

Target Theorem A – A nonconstant Euclidean action density

Rotation step: use the constant-one cone, the insertion and plate estimates, and V's two-point upper bound to eliminate nonconstant angular harmonics. **Required:** compatible global angular continuation and the polygon zero-row propagation argument, as well as the insertion estimates.

Assembly: use R's factorial bounds to extend to Schwartz tests flat on coincidence diagonals; restore the selected phase's finite mean $S_1(f) = b\int f$. Check all conclusions with the same parameters and phase.

Target: an ergodic component of a volume-first subsequential four-dimensional $SU(2)$ scalar action-density limit, with translation and O(4) invariance, reflection positivity on positive-time diagonal-flat tests, factorial growth, inverse-square clustering and connected two-point function comparable to $d^{-8}/(\log(1/d))^2$ for the fixed bump pair in the theorem.

YM4O1: full Target Theorem A; rotations, pp. 33–37; flat extension, pp. 41–43

Same position in YM3: YAO1; QED3: QEM1 / QEO1. The observable is the scalar plaquette action density.

Ochre, white boxes: model-specific tasks within YM4O1; Target Theorem A remains unproved. Joined inputs are required together; crossings without a dot do not join.

Arrows record required inputs to the proposed proof, subject to the stated open obligations. Corresponding positions indicate structural reuse, not imported theorems.

The thirteen-position layout follows QEO1 and its YM3 analogue YMO1. Links open the 44-page detailed YM4O1 of 29 October 2026.